Markscheme

Mathematics: analysis and approaches
Practice question bank

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94. (a)

The product of any two consecutive integers, $n(n+1)$, is always even, since one of the two integers must be even.

Using this result, find the smaller of two consecutive positive integers whose product is 812.

set up the equation: $n(n+1)=812$ *(M1)*

$$n^2+n-812=0$$ **A1**

solve using the quadratic formula: $n=\dfrac{-1\pm\sqrt{1+3248}}{2}=\dfrac{-1\pm 57}{2}$ *(M1)* **A1**

reject the negative root; the smaller integer is $n=28$ (check: $28\times 29=812$) **R1**

[6 marks]